



First Semester 2026-27

Course Handout Part II

Date: 1st August 2026

In addition to part I (General Handout for all courses appended to the Timetable), this portion gives further specific details regarding the course.

Course Number : ECON F 412 / FIN F 313
Course Title : Security Analysis and Portfolio Management
Instructor-in-charge : Prof. Arya Kumar
Instructor : Prof. Rajan Pandey

1. Course description: Security Analysis and Portfolio Management addresses two connected problems: what a financial asset is worth, and how much of it a particular investor should hold.

The first half of the course develops valuation. Students examine the characteristics and risk-return profiles of the major asset classes – equity, fixed income, and derivatives – and the fundamental, quantitative and technical techniques used to estimate intrinsic worth. Analysis of economy, industry, and stock specific fundamentals are carried out for valuing stocks. Hence, asset valuation forms a major component of this course.

Another important aspect of the course is on modern portfolio concepts, where the criterion changes: a holding is no longer judged on its own merits but by its contribution to the risk and return of the whole. This covers modern portfolio theory, diversification and correlation, asset allocation, and the construction of portfolios aligned to a stated investor's objectives, constraints and risk tolerance.

2. Scope and Objective: The objectives of the course are to provide an introduction to:

- Types of financial securities and their salient characteristics.
- Investment philosophy and its role in asset allocation and long-term personal finance management.
- Modern portfolio theories and their applications in portfolio design and evaluation.
- Equity and debt valuation using fundamental analysis and relative valuation techniques.
- Technical analysis.
- Financial derivative contracts and their applications.

3. Course learning outcomes:

- **Distinguish** the major classes of financial securities – equity, fixed income, derivatives and alternatives – by their claim structure, risk-return profile and role within a portfolio.
- **Value** individual securities using techniques appropriate to the instrument, including discounted cash flow and relative valuation for equity and yield-based measures for fixed income, and state the assumptions on which each valuation rests.
- **Measure and decompose** investment risk, separating systematic from unsystematic components, and evaluate investments on a risk-adjusted basis rather than on raw return.
- **Construct** a diversified portfolio using correlation and modern portfolio theory, and justify the resulting asset allocation against a specified investor's objectives, time horizon and risk tolerance.
- **Apply** derivative instruments to hedge or modify portfolio exposure, and explain the payoff profile and risk of each strategy employed.

- Assess portfolio performance using risk-adjusted measures and attribution analysis, and determine when a portfolio requires rebalancing as market conditions change.

4. Course Plan:

Lecture No.	Learning Objectives	Topics to be covered	Reading Reference	Learning Outcomes
1-7	Exposure to investment climate, risk and return quantification	Introduction to investments, risk and return, investor life cycle, securities and its various types, global security markets, organization and functioning of security markets, indexes and their calculations, types of orders, trading mechanism, common stock exchange problems, margin trading	Chap 1-2 and 4-5 and Class Notes	A. Understand the foundational concepts of investments, types of securities, and investor behavior across life stages. B. Explain the structure and functioning of global security markets, stock exchanges, and trading mechanisms. C. Analyze risk-return trade-offs, interpret index calculations, and apply basic trading concepts including order types, margin trading, and market anomalies.
8-16	Modern Portfolio Theories	Efficient Market Hypothesis, Markowitz's mean variance portfolio theory, Capital market theory, Security market line, Asset Pricing Models	Ch. 6, 7, 8, Class Notes	A. Understand and evaluate the principles of market efficiency and their implications for investment decisions. B. Apply portfolio theory to construct efficient portfolios using risk-return optimization. C. Analyze asset pricing models such as CAPM, and interpret the security market line in assessing expected returns and risk premiums.
17-22	Valuation of Financial Instruments	Analysis of Financial Statements, Security valuation principles	Ch 10, 11	A. Interpret and analyze financial statements to assess the financial health and performance of firms. B. Apply fundamental valuation techniques to estimate the intrinsic value of equity and debt securities.
23-27	Time Value of Money and Fixed Income Instruments	Time Value of Money, Bond Valuation, concept of yield to maturity, yield-price relationship	Ch 17-18 and Class Notes	A. Apply time value of money concepts to evaluate cash flows and investment decisions B. Analyze bond pricing and valuation, including the calculation and interpretation of yield to maturity. C. Understand the inverse relationship between bond prices and yields and its implications for investors.
28-35	Exposure to Fundamental Analysis and Technical analysis approach to Security Valuation	Economy, industry & company analysis, Models of security valuation (Cash Flow Model and Dividend Discount Model), Multiple Growth rates, Technical chart patterns	Chap 12, 13, 14, 16 (TB)	A. Conduct top-down analysis of the economy, industry, and individual companies to inform investment decisions B. Apply fundamental valuation models such as the Dividend Discount Model and Cash Flow Model under single and multiple growth rate assumptions. C. Interpret technical chart patterns to identify trends and support trading or portfolio strategies.
36-41	Derivatives	Introduction to Options, Futures, Forwards, derivatives trading mechanics, pay-off diagrams, application of derivatives in hedging and speculation	Chap 20-22(TB) select topics, Class Notes	A. Understand the structure and functioning of derivative instruments including options, futures, and forwards. B. Analyze pay-off diagrams and trading mechanics to evaluate derivative positions. C. Apply derivatives for hedging risk and speculation in financial decision-making.

5. Text books and references:

TB1. Reilly Frank K. and Keith C. Brown, Analysis of Investments & Management of Portfolios, 10thEd., Cengage Learning, 2012

RB1. Fisher and Jordan, Security Analysis and Portfolio Management, Pearson, 6th Ed.

RB2. Eugene F. Brigham and Michael C. Ehrhardt, Financial Management: *Theory and Practice*, 12thEd., Cengage Learning, 2013

RB3. Robert A. Strong, Portfolio Construction, Management, and Protection, 4thEd., Thomson SouthWestern, 2007

RB4. Zvi Bodie, Alex Kane, Alan J. Marcus and Pitabas Mohanty, Investments, 8thEd., Tata McGraw Hill Education, 2009

RB5. Aswath Damodaran, Damodaran on Valuation: *Security Analysis for Investment and Corporate Finance*, 2nd Ed., Wiley India, 2009

6. Evaluation Scheme: Maximum marks 300.

EC No.	Components	Duration	Weight-age (%)	Date, Time	Nature of Component
1	Mid Term Exam	90 Minutes	35	As per timetable	Closed Book
2	Quiz (Best 2 of 3)*	15 minutes	10	Unannounced (Surprise)	During the regular lecture hour.
3	Assignments/ Projects	1-3 assignments	15	TBA	Take home
4	Comprehensive Exam	180 minutes	40	As per timetable	Partially OB

*** The permission to appear for the surprise quiz is strictly based on your presence in the class on the day of the quiz. It will not be granted to such students who reach the class after the quiz has started.**

Criterion for NC: a) Scoring less than 15% of marks, b) Uninformed absence in major test components such as mid-sem test and comprehensive exam, c) Use of unfair practice or plagiarism in evaluation components such as quizzes and assignments.

7. Chamber consultation hours: TBA

8. Makeup Policy: Make-up shall be granted only in genuine cases and after obtaining prior approval from the I/C.

Instructor-in-charge

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